AI-ready metadata in practice

Lessons from building Portolan

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Who is talking

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- Twenty-plus years in the geospatial industry
- Long-time contributor to the open-source software community
- **Portolan** SDI project collaborator

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THIS TALK

slides	cayetanobv.github.io/ogc-metadata-summit-bolzano
catalog	github.com/cayetanobv/south-tyrol-geodata-portolan-mirror
project	portolan-sdi.org

What Portolan is

Portolan is an **open-source specification and toolkit** that combines **cloud-native file formats** with **clear metadata** and **documentation** so that **people and agents** can publish and use data **at any scale**.

specification and toolkit

spec v0.2.0 · `portolan` CLI · `rashid` validator · a registry of catalogs

cloud-native file formats

GeoParquet for vector · COG for raster · PMTiles for display

clear metadata

a STAC 1.1 publishing profile: what a catalog must contain

documentation

`README.md` for people · `AGENTS.md` for agents, at every level

people and agents, any scale

ten agent skills · from a city's open data to a 369 TB archive

Why Portolan: the same goals as your SDI, a lighter way to deliver them

SAME VISION · A FEDERATED SPATIAL DATA INFRASTRUCTURE, DISCOVERABLE, INTEROPERABLE, OPENLY ACCESSIBLE TO ALL

AI-ready

TODAY

SDIs were not built for agents. Agents need full data and rich metadata, fast.

INSTEAD

Agents are first-class readers.

Easy to implement

TODAY

Database, WFS, WMS, schemas, specialist staff. Most organisations cannot take part.

INSTEAD

Files in a bucket.

Scalable

TODAY

Agents crawl whole datasets. Popularity means more servers, until something breaks.

INSTEAD

Scales with nobody on call.

Low cost

TODAY

Servers always on, ops engineers, a traffic spike is an emergency.

INSTEAD

Storage and egress.

Sovereign

TODAY

Dependence on foreign clouds and vendors.

INSTEAD

The full stack in your jurisdiction.

> Two decades of SDI work got the hard part right: models, catalogues, INSPIRE themes, ISO records. What changed is the delivery layer underneath. Your catalogues and records stay authoritative.

You already know the building blocks. Most carry an OGC stamp.

Block	Role in a Portolan catalog	Status
STAC 1.1	catalog · collection · item · asset, plus the provider, table, file and web-map-links extensions	OGC Community Standard Oct 2025
GeoJSON	every STAC Item is a GeoJSON Feature	IETF RFC 7946
Cloud Optimized GeoTIFF	raster, with overviews, read by range requests	OGC Standard 2023
GeoParquet	vector: Parquet with Simple Features geometry	OGC Standards Working Group approval pending
PMTiles	tiles in one file, for map display	community format outside OGC
Zarr · **COPC**	multidimensional arrays · point clouds planned, not shipped	Zarr: OGC Community Standard 2022
HTTP range requests · **object storage** · **CORS**	the only server involved is the storage itself	IETF · W3C

Nothing here is new. Portolan uses the static side of STAC only: files, no API.

STAC describes the data. Portolan defines how it is published.

STAC 1.1.0 BASELINE + PORTOLAN v0.2.0 PROFILE

DESCRIBES + LINKS

`Catalog → Collection → Item → Asset`

same STAC objects →

PORTOLAN

v0.2.0 · pre-1.0

PROFILES THE PUBLICATION

opinionated · versioned · checkable

THE PROFILE MAKES SIX PUBLISHING CHOICES EXPLICIT

- STRUCTURE + LINKS
- FORMATS + STATISTICS
- ACCESS + HOSTING
- LICENSE + PROVENANCE
- README + AGENTS
- VISUALIZATION

schemas.portolan-sdi.org/portolan/v0.2.0/schema.json ↗

Portolan connects the pieces: publish · update · use

ONE PRACTICE ACROSS THE DATA LIFECYCLE

01 PUBLISH

- convert to open formats
- write STAC + docs
- check + upload

STATIC PUBLICATION

02 UPDATE

- change the source
- preserve provenance
- recheck + sync

STABLE LINKS

03 USE

- discover the catalog
- range-read the assets
- cite sources + limits

ONE CATALOG FOR ALL

PORTOLAN

opinionated · checkable · no new data format · nothing replaced

STAC 1.1 · GeoParquet · COG · PMTiles · HTTP · object storage

AI-READY | EASY TO IMPLEMENT | SCALABLE | LOW COST | SOVEREIGN

What Portolan adds: four things the standards leave open, made mandatory

A quality floor you can check

bbox columns and spatial order in every GeoParquet · overviews in every COG · providers and licence on every collection · range requests and CORS on the host. `rashid` reads the bytes, not just the JSON.

Two documents at every level

`README.md` for the person deciding whether to trust the data · `AGENTS.md` for the agent that needs the join key and the caveats.

No server in the request path

a folder in storage the publisher controls · cost is storage and egress · nothing runs after publishing.

Federation without a central copy

a registry of independently hosted catalogs, plain JSON in version control · producer, provider and host named on each.

DOES A STATIC MODEL SCALE? FIELDS OF THE WORLD, ONE PORTOLAN CATALOG ON SOURCE COOPERATIVE

369 TB
published

106.6 TB
served in the 28 days to 27 August 2026

0
servers in front of it

Data: Fields of the World (fieldsofthe.world), published on Source Cooperative; figures from the Portolan launch post, 2 Sep 2026.

Story 1 · A mirror of this province’s open data, published as a spec v0.2.0 catalog in five commands

```
# 1. Pull 69 of the 207 layers of South Tyrol's WFS (~12 min)
portolan extract wfs $BZ_WFS ./bz --license CC0-1.0 \
  --layers "p_bz-Health:*,p_bz-Hydrology:Watercourses,…"

# 2. Validate against the spec
portolan check
# ✗ PTL-PRV-001  no provider carries the 'producer' role  (×69)

# 3. Who produced it, who hosts it, what it is not; 11 themes
$EDITOR .portolan/metadata.yaml

# 4. Regenerate STAC + README + AGENTS.md, re-check
portolan add --force . && portolan readme
portolan check --fix && rashid check . --schema --data
# 0 error(s), 0 warning(s) across 81 files

# 5. Push to a public bucket (Milan). Nothing else runs.
portolan push gs://south-tyrol-geodata-portolan-mirror/
rashid check . --live --live-base-url $PUBLIC_URL
# 0 error(s): range requests, CORS, sizes honoured
```

The service seeded what it could. Titles in three languages, keywords, abstracts, styles from the WMS. 69 layers to GeoParquet, spatially ordered, bbox column added; PMTiles and thumbnails alongside.

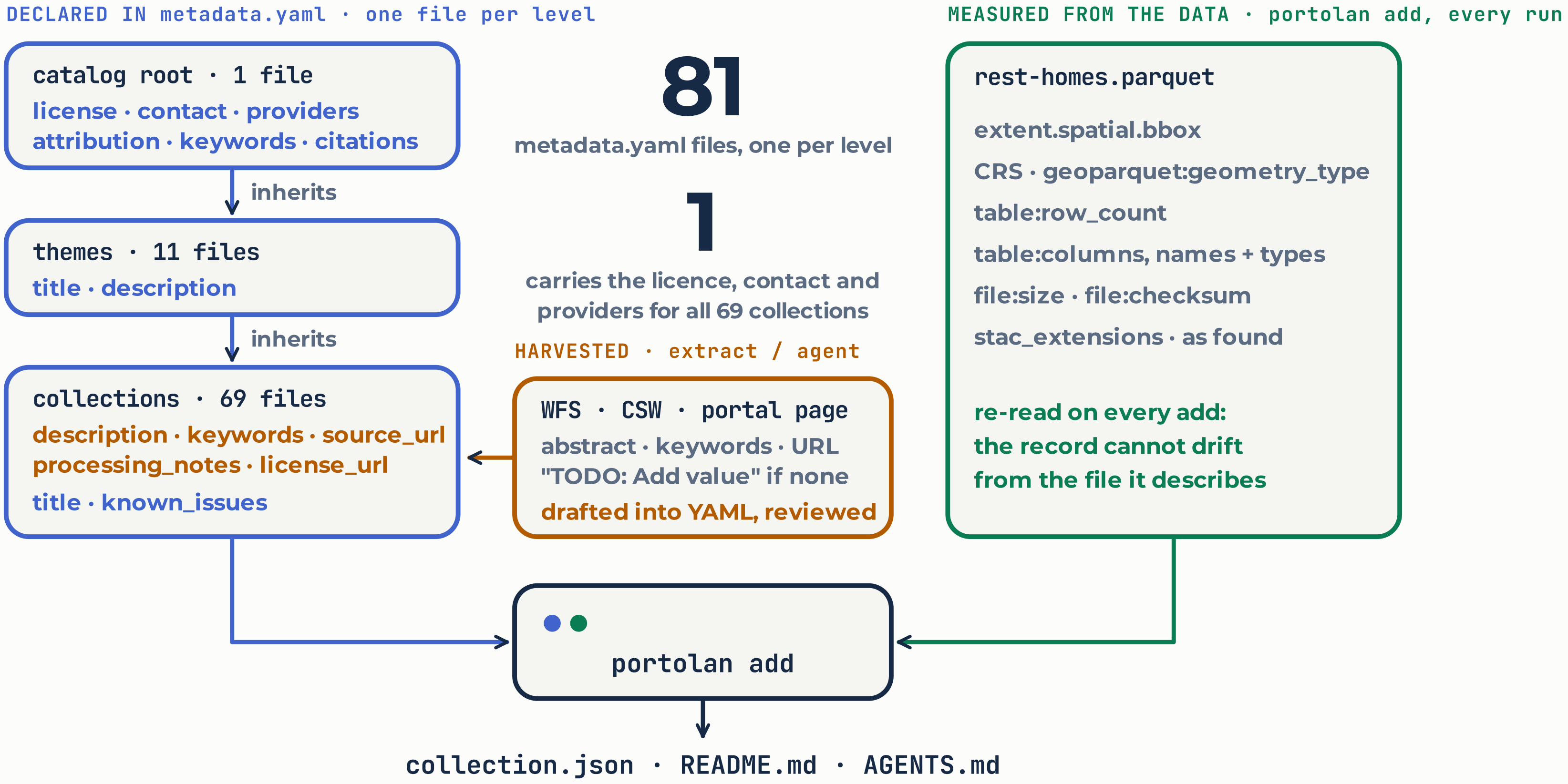
The validator failed on the machine contract, not on the prose: no producer, no license from the service. The producer's knowledge went into YAML at every level; from it came `README.md`, `collection.json`, `AGENTS.md`.

Two validators, same answer: conforms to v0.2.0, hosting contract included once pushed (range requests, CORS, sizes). The same run, hands off: an agent with the `portolan-bootstrap` skill does steps 1–5 and writes the documentation. No server was started. The result is a folder in a bucket.

EXAMPLE CATALOG open it in the Portolan browser ↗ · catalog.json on storage.googleapis.com ↗ · metadata and CI on GitHub ↗

Data: Provincia Autonoma di Bolzano – Alto Adige / Autonome Provinz Bozen – Südtirol (Ripartizione 28 · ASTAT · Ripartizione 23 Salute), CC0 1.0, via data.civis.bz.it and the provincial WFS. Community mirror extracted 5 September 2026 — not an official publication; the authoritative source is the Province's geoportal.

Where the metadata comes from: measured, harvested, asserted



drafted by a tool · approved by a person · gated by `rashid check` in CI on every push

Story 2 · Then an agent was asked a question no single layer answers

> *"Which hospitals and pharmacies in South Tyrol sit inside high or very-high hydraulic hazard zones of the approved hazard-zone plans, and how many residents live in those municipalities?"*

4 of 17
hospitals inside H3/H4 zones · Bolzano, Bressanone, Brunico, Ortisei

31 of 158
pharmacies inside H3/H4 zones · Merano 3, Bolzano 2, Bressanone 2, Brunico 2 ...

108 of 116
municipalities with an approved hydraulic hazard plan · 536,933 residents in the province

STATED BY THE AGENT, BECAUSE THE PUBLISHER WROTE IT DOWN

Points are locations, not service catchments. Hazard classes are planning zones, not event forecasts. Eight municipalities have no approved plan yet. Producer: Autonomous Province of Bolzano – South Tyrol · CC0-1.0 · query attached.

WHAT IT HAD TO WORK WITH

- 69 collections in 11 themes, `AGENTS.md` at every level
- the join key `ISTAT_CODE`, the CRS, the H2–H4 class strings, the caveats
- GeoParquet with bbox columns, read in place with DuckDB
- the `reading-portolan` skill: how to navigate, query, join and report

WHAT IT DID NOT HAVE

- a server, an API, credentials, or a person in the loop

A screening, not a risk assessment. The same catalog answers the next question tomorrow.

Data: Provincia Autonoma di Bolzano – Alto Adige — Comuni · Popolazione ufficiale · Piani delle zone di pericolo: pericoli idraulici · Ospedali · Farmacie — CC0 1.0, data.civis.bz.it. Run on a community mirror; a screening, not an official assessment.

What the agent actually did with the metadata

1

Open `catalog.json`

follow the explicit child links: 11 theme catalogs, 69 collections; the linked graph is the publication boundary

2

Read `AGENTS.md`

join key `ISTAT_CODE` , CRS, the H2–H4 class strings, what the data is not for

3

Inspect each collection

picks five of the 69: one GeoParquet asset each, with provider, license, extent, row count

4

Plan a DuckDB query

the model plans and explains; DuckDB computes, reading only the row groups it needs

5

Report

method, exact URLs and query, producer, license, limitations

Every step consumed something the publisher wrote down. Nothing was guessed. That is the operational meaning of AI-ready metadata.

LIVE AT THE DEMO DESK

This story, live: ask the agent *your* question on the South Tyrol catalog, and watch it read the metadata before it answers.

Recording of this run, 1 min 33 s ↗

The skills: the know-how that made both stories run

STORY 1 · PUBLISHING

- `portolan-bootstrap` — from a WFS, an Esri service, a portal or a folder to a documented, validated, published catalog
- `portolan-cli` — init, add, check, push, sync, conversion
- `portolan-thumbnails` — framed, checked thumbnails over a basemap
- `portolan-migrate` — bring an existing catalog up to the spec without rebuilding it
- `git-backed-catalog` · `sourcecoop` · `register-catalog` — versioned metadata, hosting, the registry pull request

STORY 2 · CONSUMING

- `reading-portolan` — the analyst's skill: navigate the STAC tree, query GeoParquet with DuckDB, join across collections, map it with PMTiles, MapLibre or deck.gl, report with sources
- `portolan-consume` — the on-ramp: detects what the user has installed (DuckDB, GeoPandas, rioxarray, rasterio) and shows how to read the GeoParquet and COG assets with those tools
- `report-catalog-issue` — a wrong licence or a broken link becomes an issue on the registry, from the agent

> Plain `SKILL.md` files in the open **Agent Skills** format: the same know-how works in Claude Code, Gemini CLI and other agents. Ten skills, one repository, Apache-2.0. **The spec says what a good catalog is; the skills encode how to build and use one.**

Lesson 1 of 3 · Discovery is the solved half. Evaluability is the gap.

WHAT A GOOD STAC CATALOG ALREADY GIVES YOU

Found. Crawled. Opened.

WHAT PEOPLE AND AGENTS STILL LACK

Is this dataset *for* my question? Where is it weak? How wrong can it be?

ONE SENTENCE FROM A REAL CATALOG (FIELDS OF THE WORLD)

A field polygon "is a *remote-sensing field unit*, **not** a cadastral/legal parcel. This is not a land-tenure product."

No accuracy figure prevents that misuse. One sentence of scope does.

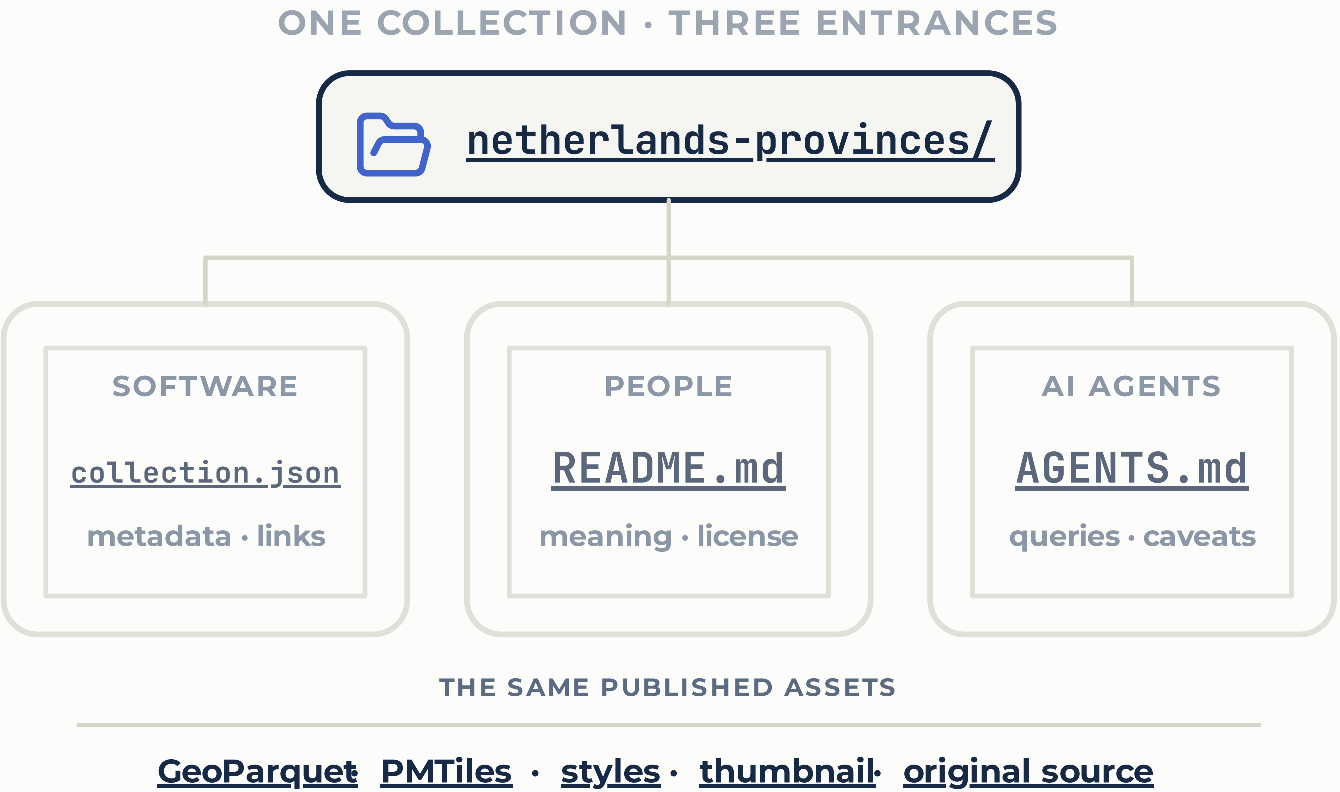
Source: Fields of the World, global field-boundary dataset documentation (fieldsofthe.world, CC BY 4.0), as quoted in the Portolan documentation best practices.

AI-ready = documented limitations, quantified uncertainty, named biases — **as data where possible** (a confidence column, a quality mask), not buried in prose. Expertise that otherwise lives only in the heads of domain experts.

WHERE A LIMITATION LIVES IN A PORTOLAN CATALOG

`known_issues:` in `metadata.yaml`, written once by the publisher → `## Known Issues` in `README.md`, for people → `## Data quality & usage notes` in `AGENTS.md`, for agents → `confidence` column as data, where the limitation is per row

Lesson 2 of 3 · Two audiences, two files, one subject



`README.md` **helps a person decide whether to trust the data.** `AGENTS.md` **helps an agent that already committed to using it get the query right the first time. Both generated from one source.**

Lesson 3 of 3 · Be strict, and let a machine hold the line

WHY SO PRESCRIPTIVE

An agent can only rely on what is *required*. Every MAY in a specification is a branch every consumer has to write.

- GeoParquet without bbox columns and spatial ordering: spatial queries read the whole file
- a bucket without CORS: unusable from a browser
- an empty `providers` list: the prose looks fine, every machine fails

WHAT THE MACHINE HOLDS · SPEC v0.2.0

128 normative requirements

105 checked deterministically by `rashid`

23 stay with the publisher's judgment

MUST → failure, SHOULD → warning, JSON out. A city's catalog passes the same validator a planetary archive passes.

Conformance is a floor. "You can have perfectly FAIR, utterly useless data." Above pass/fail, a grader scores usability from A+ to F, and weights the machine contract equally with what a browser shows.

A metadata quality policy only sticks when a machine can check it.

Honestly unfinished: trust and provenance

How should **authority** be signalled cryptographically, not just declared in a providers list?

What does **trust** mean when the consumer is an agent acting unattended?

How does a **mirror** stay verifiably faithful to its source?

Today: every collection names producer, licensor, processor and host; official or mirror is derived from that; a registry of ~20 independently hosted catalogs, plain JSON in git. All declarative. The questions above map onto the OGC Testbed on Trusted Data and Systems.

Public since 2 September 2026. Early adopters wanted; breaking changes still expected. First goal: ~100 reference catalogs, then a stable v1.0 CLI.

Questions? Everything is open and one pull request away.

PROJECT LINKS

CLICK ANY CARD

WEBSITE
portolan-sdi.org ↗

ALL PROJECTS
github.com/portolan-sdi ↗

SPECIFICATION
portolan-spec ↗

PUBLISHER CLI
portolan-cli ↗

VALIDATOR
rashid ↗

AGENT SKILLS
portolan-skills ↗

PUBLIC REGISTRY
portolan-registry ↗

CATALOG BROWSER
browser.portolan-sdi.org ↗

THIS DECK

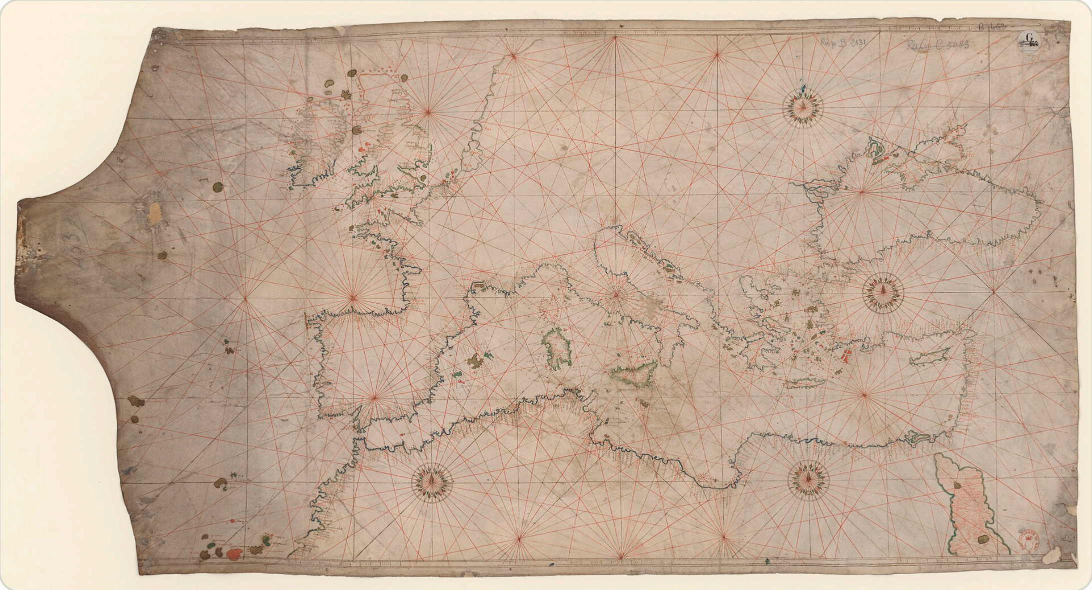
cayetanobv.github.io

ogc-metadata-summit-bolzano/#/18

SCAN · CLICK · SHARE

COMMUNITY #portolan on the Cloud-Native Geospatial Forum Slack

Appendix 1 · Why "Portolan"



Portolan chart of the Mediterranean and the Black Sea, Battista Agnese, 1550. Bibliothèque nationale de France, via Wikimedia Commons (public domain).

Portolan charts were among the earliest **practical** navigational maps: not theoretical pictures of the world but working tools, built from sailors' observations, continuously refined, and shared across ports and nations.

- **Decentralised** — made and improved by many hands
- **Interoperable** — readable wherever a ship went
- **Critical infrastructure** — for trade, for safety, for sovereignty

> **The project takes its name from that tradition. Practical rather than academic; built from data in open formats rather than drawings; designed for sharing across organisations, clouds and borders.**

Appendix 2 · ISO 19115 / INSPIRE records as inputs

ISO IN

`portolan extract wfs` reads the ISO 19139 record over CSW and seeds title, abstract, keywords, license, contact, lineage, dates.

ISO ALONGSIDE

The record travels with the data as an asset with `roles: ["metadata","iso-19115"]` and a typed `describedby` link to the catalogue that owns it.

GEODCAT-AP OUT

Generated from the same source as a sibling representation, so open-data portals can harvest it. Roadmap, not shipped.

The same pattern swisstopo and the CEOS STAC best practices already follow. There is still no ISO 19115 STAC extension — an open question for this room.

Appendix 2 · How an ISO record rides in a Portolan collection

```
"assets": {
  "metadata": {
    "href": "./iso19115.xml",
    "type": "application/vnd.iso.19139+xml",
    "title": "ISO 19115 record, BRK Bestuurlijke Gebieden (Nationaal Georegister)",
    "roles": ["metadata", "iso-19115"],
    "file:size": 40942, "file:checksum": "1220…"
  }
},
"links": [
  {"rel": "describedby", "href": "./README.md", "type": "text/markdown"},
  {"rel": "describedby",
   "href": "https://www.nationaalgeoregister.nl/geonetwork/srv/api/records/208bc283-…/formatters/xml",
   "type": "application/vnd.iso.19139+xml", "title": "ISO 19115 record — authoritative"},
  {"rel": "via", "href": "https://www.pdok.nl/…", "type": "text/html", "title": "Original source (PDOK)"}
]
```

Asset = pinned copy with checksum and mirror date. Link = live authoritative record. If they disagree, the link wins and the asset tells you how old the copy is.